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Course : Data Warehousing And Data Mining

Course code : CSA1606

Experiment 1

Aim

To build a WEKA application using the J48 Decision Tree algorithm to predict whether a person should play outdoors based on weather conditions.

Algorithm

J48 Decision Tree

Input

Outlook

Temperature

Humidity

Windy

Output

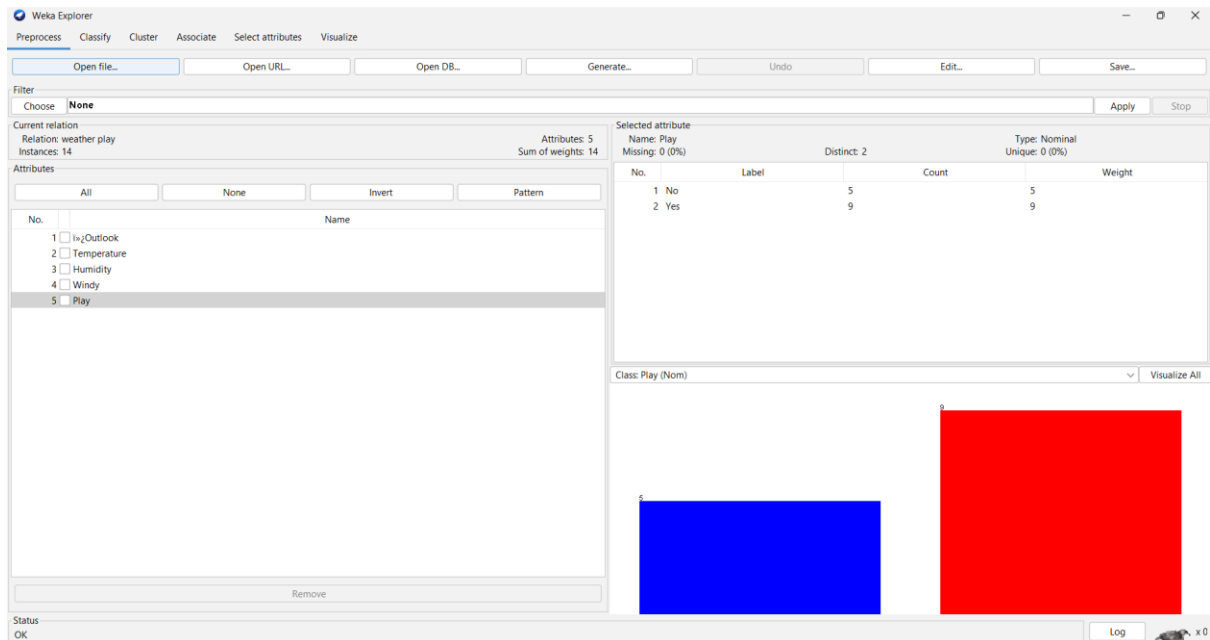
Play = Yes / No

Procedure

1. Open WEKA Explorer.
2. Create the weather dataset in CSV format.
3. Load the dataset using the Preprocess tab.
4. Select Play as the class attribute.
5. Go to the Classify tab.
6. Select J48 from Trees.
7. Select 10-fold cross-validation.
8. Click Start.
9. Observe the classification accuracy.

10. Right-click the result and select Visualize tree.

11. Provide a new weather condition and predict the Play value.



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply Stop

Current relation: Relation: weather play Instances: 14

Attributes: 5 Sum of weights: 14

Attributes: All None Invert Pattern

No. Name

1 ☐ Outlook

2 ☐ Temperature

3 ☐ Humidity

4 ☐ Windy

5 ☒ Play

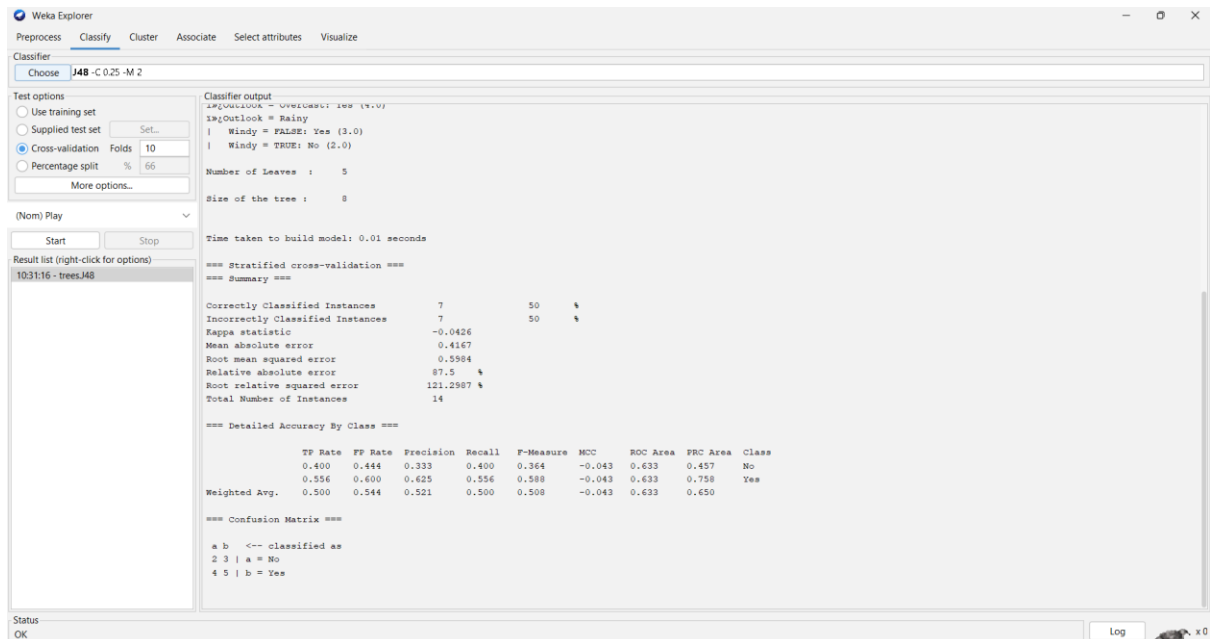
Remove

Status: OK Log

Selected attribute: Name: Play Missing: 0 (0%) Distinct: 2 Type: Nominal Unique: 0 (0%)

No.	Label	Count	Weight
1	No	5	5
2	Yes	9	9

Class: Play (Nom) Visualize All



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **J48 - C 0.25 - M 2**

Test options:

☐ Use training set

☐ Supplied test set Set...

☒ Cross-validation Folds: 10

☐ Percentage split %: 66

More options...

(Nom) Play

Start Stop

Result list (right-click for options): 10:31:16 - treesJ48

Classifier output:

```
1>Outlook = Overcast: yes (4.0)
1>Outlook = Rainy
| Windy = FALSE: Yes (3.0)
| Windy = TRUE: No (2.0)

Number of Leaves : 5
Size of the tree : 8

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===
=== Summary ===

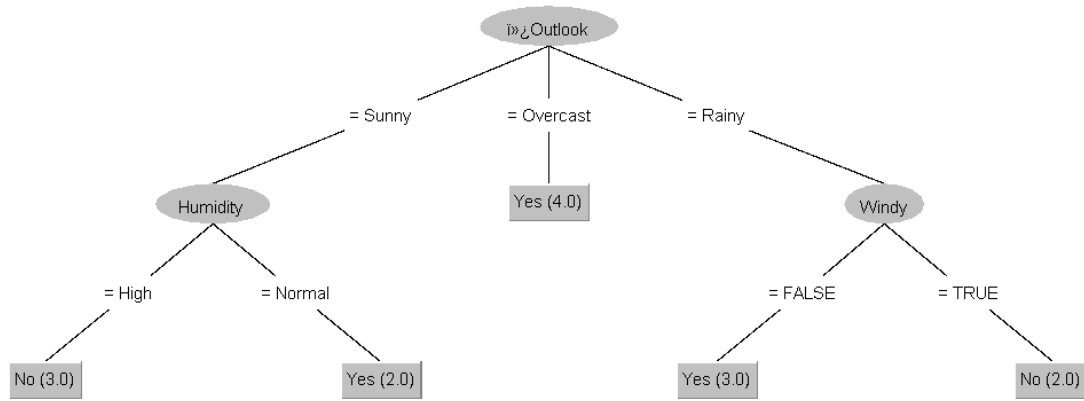
Correctly Classified Instances 7 50 %
Incorrectly Classified Instances 7 50 %
Kappa statistic -0.0426
Mean absolute error 0.4167
Root mean squared error 0.5904
Relative absolute error 87.5 %
Root relative squared error 121.2907 %
Total Number of Instances 14

=== Detailed Accuracy By Class ===

TP Rate FP Rate Precision Recall F-Measure MCC ROC Area SPC Area Class
0.400 0.444 0.333 0.400 0.364 -0.043 0.633 0.457 No
0.556 0.600 0.625 0.556 0.588 -0.043 0.633 0.758 Yes
Weighted Avg. 0.500 0.544 0.521 0.500 0.508 -0.043 0.633 0.650

=== Confusion Matrix ===
a b <-- classified as
2 3 | a = No
4 5 | b = Yes
```

Status: OK Log



Experiment 2

Aim

To develop a WEKA application using Linear Regression to predict temperature based on the number of days.

Algorithm

Linear Regression

Input

Day

Output

Temperature

Procedure

1. Create the temperature dataset in CSV format.
2. Open WEKA Explorer.
3. Load the dataset using the Preprocess tab.
4. Select Temperature as the class attribute.
5. Go to the Classify tab.

6. Select LinearRegression.
7. Select the required evaluation method.
8. Click Start.
9. Observe the regression equation and evaluation results.
10. Use the regression equation to predict temperature for Day 10.
11. Calculate the temperature increase.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply Stop

Current relation: weather prediction
Instances: 7

Attributes: 2
Sum of weights: 7

Selected attribute: Name: Temperature
Missing: 0 (0%)
Distinct: 7
Type: Numeric
Unique: 7 (100%)

Statistic	Value
Minimum	25
Maximum	31
Mean	28
StdDev	2.16

Class: Temperature (Num) Visualize All

Status: OK

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4**

Test options:

- ☒ Use training set
- ☐ Supplied test set
- ☐ Cross-validation
- ☐ Percentage split

Folds: 10
Percentage split: 66

More options...

(Num) Temperature

Start Stop

Result list (right-click for options): 10:45:12 - functions.LinearRegression

Classifier output:

```

=== Run information ===

Scheme: weka.classifiers.functions.LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4
Relation: weather prediction
Instances: 7
Attributes: 2
  1%Day
  Temperature
Test mode: evaluate on training data

=== Classifier model (full training set) ===

Linear Regression Model
Temperature =
  1 * 1%Day +
  24

Time taken to build model: 0.06 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correlation coefficient      1
Mean absolute error         0
Root mean squared error     0
Relative absolute error     0
Root relative squared error  0
Total Number of Instances   7
  
```

Status: OK